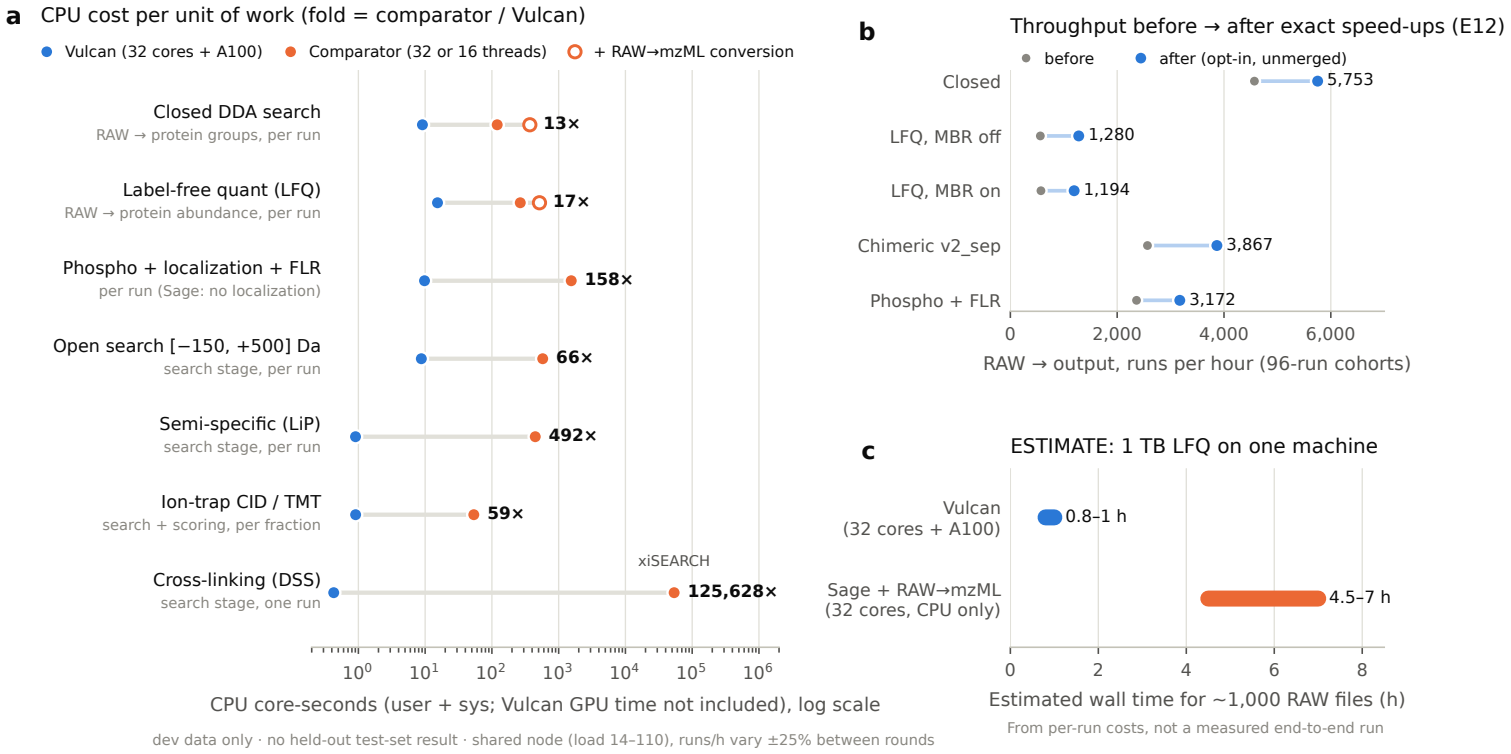
a
CPU cost per unit of work (fold = comparator / Vulcan)
Vulcan (32 cores + A100)
Comparator (32 or 16 threads)
+ RAW→mzML conversion
Closed DDA search
RAW → protein groups, per run
13×
Label-free quant (LFQ)
RAW → protein abundance, per run
17×
Phospho + localization + FLR
per run (Sage: no localization)
158×
Open search [−150, +500] Da
search stage, per run
66×
Semi-specific (LiP)
search stage, per run
492×
Ion-trap CID / TMT
search + scoring, per fraction
59×
xiSEARCH
Cross-linking (DSS)
search stage, one run
125,628×
10^0 10^1 10^2 10^3 10^4 10^5 10^6
CPU core-seconds (user + sys; Vulcan GPU time not included), log scale
b
Throughput before → after exact speed-ups (E12)
before
after (opt-in, unmerged)
Closed
5,753
LFQ, MBR off
1,280
LFQ, MBR on
1,194
Chimeric v2_sep
3,867
Phospho + FLR
3,172
0
2,000
4,000
6,000
RAW → output, runs per hour (96-run cohorts)
c
ESTIMATE: 1 TB LFQ on one machine
Vulcan
(32 cores + A100)
0.8–1 h
Sage + RAW→mzML
(32 cores, CPU only)
4.5–7 h
0
2
4
6
8
Estimated wall time for ~1,000 RAW files (h)
From per-run costs, not a measured end-to-end run
dev data only · no held-out test-set result · shared node (load 14–110), runs/h vary ±25% between rounds